MATH 423 Introduction to Computational Topology

Lecture Notes on Simplicial Homology

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Abstract

These notes collect the material on simplicial homology covered before the first midterm: chains and boundary operators over $\mathbb{Z}/2$, cycles, boundaries and Betti numbers, the computation of homology by row reduction, the meaning of H_0, the Euler–Poincaré formula, cones and one-point unions, the homology of Čech and Vietoris–Rips complexes, integer coefficients and the first homology of surfaces, functoriality, and local homology. Each section ends with worked examples; a list of computational recipes and exercises with answers is at the end.

Midterm 1 covers Sections 1–8, with all homology taken over $\mathbb{Z}/2$. Sections 9–11 (integer coefficients, invariance and functoriality, local homology) and the exercises marked **[Not on Midterm 1]** are not on Midterm 1; they will be used later in the course.

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1 Review and Motivation

1.1 Simplicial complexes and the Euler characteristic

An **abstract simplicial complex** K on a finite vertex set V is a collection of nonempty subsets of V, the **simplices**, such that every singleton $\{v\}$ is in K and every nonempty subset (**face**) of a simplex is again a simplex. A simplex with $k+1$ vertices has **dimension** k; we write n_k for the number of k-simplices. The f**-vector** of K is $(n_0, n_1, \ldots, n_d)$, where $d = \dim K$ is the largest dimension of a simplex, and the **Euler characteristic** is

$$\chi(K) = n_0 - n_1 + n_2 - \cdots + (-1)^d n_d.$$

A complex is usually given by its **maximal simplices**, those that are not faces of larger ones; all other simplices are obtained by taking faces. The k**-skeleton** $K^{(k)}$ consists of the simplices of dimension at most k.

The Euler characteristic is a topological invariant, but it is a single number that mixes information from all dimensions. A 2-sphere and a space consisting of two points both have $\chi = 2$. Homology refines χ into a sequence of numbers, the Betti numbers $b_0, b_1, b_2, \ldots$, one for each dimension, with $\chi = b_0 - b_1 + b_2 - \ldots$ (Theorem 6.1). For the sphere we will find $(b_0, b_1, b_2) = (1, 0, 1)$ and for two points $(2, 0, 0)$.

1.2 Stars and links

Let σ be a simplex of K.

- The **star** $\mathrm{St}(\sigma)$ is the set of simplices of K that contain σ.
- The **closed star** $\overline{\mathrm{St}}(\sigma)$ is the set of all faces of simplices in $\mathrm{St}(\sigma)$; it is the smallest subcomplex containing $\mathrm{St}(\sigma)$.
- The **link** $\mathrm{lk}(\sigma)$ is the set of simplices $\tau \in \overline{\mathrm{St}}(\sigma)$ with $\tau \cap \sigma = \emptyset$; equivalently, $\tau \in \mathrm{lk}(\sigma)$ if and only if $\tau \cap \sigma = \emptyset$ and $\tau \cup \sigma \in K$.

The closed star and the link are subcomplexes. The star is in general *not* a subcomplex: if K is a single edge $\{v, w\}$ with its two vertices, then $\mathrm{St}(v) = \{\{v\}, \{v, w\}\}$ contains the edge but not its face $\{w\}$. For a vertex v in a triangulated surface, $\mathrm{lk}(v)$ is a circle (or a path, if v is on the boundary); this will be the starting point of local homology in Section 11.

1.3 Čech and Vietoris–Rips complexes

Let X be a finite subset of a metric space (M, d) and $\epsilon > 0$. We use the following conventions.

- $\sigma \subseteq X$ is a simplex of the **Čech complex** $\check{C}_\epsilon(X)$ if the closed balls of radius ϵ centered at the points of σ have a common point.
- $\sigma \subseteq X$ is a simplex of the **Vietoris–Rips complex** $VR_\epsilon(X)$ if $d(x, y) \le 2\epsilon$ for all $x, y \in \sigma$.

Both are simplicial complexes on the vertex set X, and both grow with ϵ. The Vietoris–Rips complex is a **flag complex**: it is determined by its edges, since a set of vertices is a simplex as soon as all its pairs are edges.

Proposition 1.1. *For every finite X and every $\epsilon > 0$, $\check{C}_\epsilon(X) \subseteq VR_\epsilon(X)$. If $X \subset \mathbb{R}^2$, then also $VR_\epsilon(X) \subseteq \check{C}_{\sqrt{2}\epsilon}(X)$.*

Proof. If $\sigma \in \check{C}_\epsilon(X)$, there is a point p with $d(x, p) \le \epsilon$ for all $x \in \sigma$, so $d(x, y) \le d(x, p) + d(p, y) \le 2\epsilon$ for $x, y \in \sigma$.

For the second inclusion, let $\sigma \in VR_\epsilon(X)$, $X \subset \mathbb{R}^2$, and consider the smallest closed disk containing σ. It is determined either by two points of σ lying on a diameter, in which case its radius is at most $\frac{1}{2} \cdot 2\epsilon = \epsilon$, or by three points of σ forming a triangle with no obtuse angle, in which case its radius is the circumradius $R = a/(2 \sin \alpha)$, where $a \le 2\epsilon$ is the longest side and α the largest angle. Since $60^\circ \le \alpha \le 90^\circ$, we get $\sin \alpha \ge \sqrt{3}/2$ and $R \le 2\epsilon/\sqrt{3} < \sqrt{2}\epsilon$. The center of this disk lies in every ball of radius $\sqrt{2}\epsilon$ around a point of σ. □

The Čech complex is the nerve of the union of balls, so by the Nerve Theorem it has the same homotopy type, and hence the same homology (Theorem 10.1), as $\bigcup_{x \in X} B(x, \epsilon)$. The Vietoris–Rips complex is easier to compute, since only pairwise distances are needed, but it does not have this property; see Example 8.1.

2 Chains and the Boundary Operator over $\mathbb{Z}/2$

Throughout, K is a finite simplicial complex. We begin with coefficients in the field $\mathbb{Z}/2 = \{0, 1\}$, where $1 + 1 = 0$. This is the setting used by most software for topological data analysis, and it has the advantage that no orientations are needed. Integer coefficients are introduced in Section 9.

Definition 2.1. A **k-chain** of K with coefficients in $\mathbb{Z}/2$ is a formal sum

$$c = \sum_{\sigma} a_\sigma \sigma, \qquad a_\sigma \in \mathbb{Z}/2,$$

over the k-simplices σ of K. The k-chains form a vector space $C_k(K; \mathbb{Z}/2)$ over $\mathbb{Z}/2$, with the k-simplices as a basis, so $\dim C_k = n_k$. We set $C_k = 0$ for $k < 0$ and for $k > \dim K$.

Since each coefficient is 0 or 1, a k-chain is the same as a *set* of k-simplices, namely those with coefficient 1. Adding two chains corresponds to the symmetric difference of the sets: a simplex appearing in both chains gets coefficient $1 + 1 = 0$ and disappears. In a computer, a k-chain is a vector of n_k bits.

Definition 2.2. The **boundary** of a k-simplex $\sigma = \{v_0, \dots, v_k\}$ is the sum of its $(k-1)$-dimensional faces,

$$\partial_k \sigma = \sum_{i=0}^{k} \{v_0, \dots, \widehat{v_i}, \dots, v_k\},$$

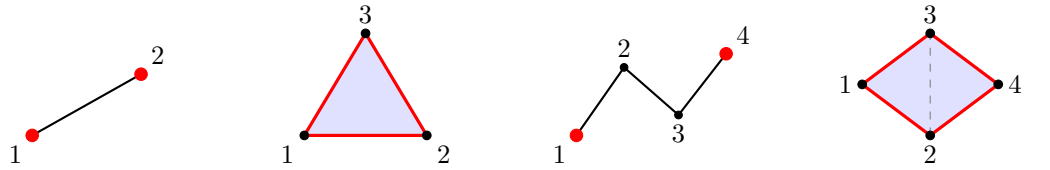
where the hat means that v_i is omitted. This extends linearly to the **boundary operator** $\partial_k : C_k(K; \mathbb{Z}/2) \to C_{k-1}(K; \mathbb{Z}/2)$. For $k = 0$ we set $\partial_0 = 0$.

Example 2.3.

- The boundary of an edge is its two endpoints: $\partial_1\{1,2\} = \{1\} + \{2\}$.
- The boundary of a triangle is its three edges: $\partial_2\{1,2,3\} = \{2,3\} + \{1,3\} + \{1,2\}$.
- For a path of edges $c = \{1,2\} + \{2,3\} + \{3,4\}$,

  $$\partial_1 c = (\{1\} + \{2\}) + (\{2\} + \{3\}) + (\{3\} + \{4\}) = \{1\} + \{4\},$$

  since each interior vertex appears twice and $1 + 1 = 0$.
- For two triangles sharing an edge, $\partial_2(\{1,2,3\} + \{2,3,4\}) = \{1,2\} + \{1,3\} + \{2,4\} + \{3,4\}$: the shared edge $\{2,3\}$ cancels, and what remains is the boundary of the square formed by the two triangles.



(a) an edge: $\{1\} + \{2\}$ (b) a triangle: its three edges (c) a path: $\{1\} + \{4\}$ (d) two triangles: shared edge cancels

Figure 1: The boundary operator over $\mathbb{Z}/2$; boundaries are drawn in red. In (c) each interior vertex of the path is counted twice and cancels. In (d) the edge $\{2,3\}$ (dashed) is a face of both triangles and cancels.

With respect to the bases of simplices, ∂_k is given by an $n_{k-1} \times n_k$ matrix with entries in $\mathbb{Z}/2$, the **boundary matrix**. Its rows are indexed by the $(k-1)$-simplices and its columns by the k-simplices; the column of σ has a 1 in the row of each facet of σ, so every column of ∂_k contains exactly $k+1$ ones.

Example 2.4. For the hollow triangle with vertices $1, 2, 3$ and edges $\{1,2\}$, $\{1,3\}$, $\{2,3\}$, in this order,

$$\partial_1 = \begin{array}{c|ccc} & 12 & 13 & 23 \\ \hline 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ 3 & 0 & 1 & 1 \end{array}$$

The following identity is the foundation of everything that follows.

Lemma 2.5 (The boundary of a boundary is zero)**.** *For every k, $\partial_{k-1} \circ \partial_k = 0$.*

Proof. By linearity it suffices to check this on a single k-simplex $\sigma = \{v_0, \dots, v_k\}$. Applying ∂ twice gives the sum of all the faces of σ obtained by deleting two vertices. Each such face $\sigma \setminus \{v_i, v_j\}$ with $i < j$ arises exactly twice, once by deleting v_i first and once by deleting v_j first. Over $\mathbb{Z}/2$ each face therefore appears with coefficient $1 + 1 = 0$. □

For example, $\partial_1 \partial_2 \{1,2,3\} = (\{2\} + \{3\}) + (\{1\} + \{3\}) + (\{1\} + \{2\}) = 0$. Geometrically: the boundary of a triangle is a closed loop, and a closed loop has no endpoints. In matrix form the lemma says $\partial_{k-1}\partial_k = 0$ as a product of matrices, which is a useful check on hand computations.

3 Cycles, Boundaries and Homology

Definition 3.1.

- A k-chain c is a **k-cycle** if $\partial_k c = 0$. The k-cycles form the subspace $Z_k = \ker \partial_k$ of C_k.
- A k-chain c is a **k-boundary** if $c = \partial_{k+1} d$ for some $(k+1)$-chain d. The k-boundaries form the subspace $B_k = \operatorname{im} \partial_{k+1}$ of C_k.

By Lemma 2.5, every boundary is a cycle: $B_k \subseteq Z_k$. The **k-th homology group** of K with coefficients in $\mathbb{Z}/2$ is the quotient vector space

$$H_k(K; \mathbb{Z}/2) = Z_k / B_k,$$

and its dimension $b_k = \dim H_k(K; \mathbb{Z}/2)$ is the **k-th Betti number** of K over $\mathbb{Z}/2$.

Two cycles z, z' define the same element of H_k if $z - z' = z + z'$ is a boundary; we then say that z and z' are **homologous** and write $[z] = [z']$. Intuitively, a 1-cycle is a collection of closed loops (a set of edges meeting every vertex an even number of times), and it is a boundary if it can be filled in by triangles of K. So H_1 measures the loops that cannot be filled in, that is, the 1-dimensional holes. Similarly, a 2-cycle is a closed surface made of triangles of K, and it is a boundary if it can be filled in by tetrahedra.

Since $\dim Z_k = n_k - \operatorname{rank} \partial_k$ by the rank–nullity theorem and $\dim B_k = \operatorname{rank} \partial_{k+1}$, the Betti numbers are given by

$$\boxed{b_k = n_k - \operatorname{rank} \partial_k - \operatorname{rank} \partial_{k+1}}$$

where all ranks are computed over $\mathbb{Z}/2$. Computing homology is therefore reduced to computing ranks of boundary matrices. Note that b_k only involves ∂_k and ∂_{k+1}, so it depends only on the $(k+1)$-skeleton of K: to compute b_1 one needs vertices, edges and triangles, but no tetrahedra.

Example 3.2 (A point)**.** If $K = \{\{v\}\}$, then $C_0 = \mathbb{Z}/2$ and all other chain groups are zero. Since $\partial_0 = 0$, we get $Z_0 = C_0$ and $B_0 = 0$, so $b_0 = 1$ and $b_k = 0$ for $k \geq 1$.

Example 3.3 (The hollow triangle)**.** For the hollow triangle, $n_0 = 3$ and $n_1 = 3$. The three rows of ∂_1 add up to zero, and any two of them are independent, so $\operatorname{rank} \partial_1 = 2$. There are no triangles, so $\partial_2 = 0$. Hence

$$b_0 = 3 - 0 - 2 = 1, \qquad b_1 = 3 - 2 - 0 = 1.$$

The cycle space Z_1 is spanned by $z = \{1,2\} + \{1,3\} + \{2,3\}$, and since there is no triangle, z is not a boundary. There is one hole, as expected for a circle.

Example 3.4 (The solid triangle)**.** Adding the triangle $\{1,2,3\}$ gives $n_2 = 1$ and $\partial_2\{1,2,3\} = z$. Now z is a boundary, $\operatorname{rank}\partial_2 = 1$, and

$$b_0 = 1, \qquad b_1 = 3 - 2 - 1 = 0, \qquad b_2 = 1 - 1 - 0 = 0.$$

Filling in the triangle has killed the hole; the solid triangle has the homology of a point.



z is a cycle, but not a boundary: $b_1 = 1$

$z = \partial_2\{1,2,3\}$ is a boundary: $b_1 = 0$

Figure 2: The same 1-cycle $z = \{1,2\}+\{1,3\}+\{2,3\}$ in the hollow triangle and in the solid triangle. Homology counts cycles that cannot be filled in.

Example 3.5 (The sphere)**.** Let K be the boundary of the tetrahedron on $\{1,2,3,4\}$, with f-vector $(4,6,4)$. Its 1-skeleton is connected, so (Theorem 5.1) $\operatorname{rank}\partial_1 = 4 - 1 = 3$. The sum of all four triangles is a 2-cycle, since every edge lies in exactly two triangles and so cancels. No nonempty proper set of triangles is a cycle: if a 2-chain contains a triangle, it must also contain the other triangle through each of its edges, and then all four. So $\dim Z_2 = 1$ and $\operatorname{rank}\partial_2 = 4 - 1 = 3$. Therefore

$$b_0 = 4 - 0 - 3 = 1, \qquad b_1 = 6 - 3 - 3 = 0, \qquad b_2 = 4 - 3 - 0 = 1.$$

The 2-cycle formed by all four triangles encloses the 2-dimensional void of the sphere.

Example 3.6 (A graph)**.** If K is a graph, that is, a 1-dimensional complex, then $\partial_2 = 0$, so $b_1 = \dim Z_1 = n_1 - \operatorname{rank}\partial_1$. For a connected graph, $\operatorname{rank}\partial_1 = n_0 - 1$ (Theorem 5.1), so

$$b_1 = E - V + 1,$$

the number of independent cycles. For the complete graph K_4 (the 1-skeleton of the tetrahedron), $b_1 = 6 - 4 + 1 = 3$; the boundaries of three of the four triangles form a basis of Z_1, and the fourth is their sum.

3.1 Deciding whether a cycle is a boundary

Two kinds of questions come up constantly: *is this chain a cycle*, *is this cycle a boundary*, and *are these two cycles homologous*. Over $\mathbb{Z}/2$ they are answered as follows.

- **Cycle.** A 1-chain is a cycle if and only if every vertex lies on an even number of its edges. A 2-chain is a cycle if and only if every edge lies in an even number of its triangles.
- **Homologous.** To show $[z] = [z']$, exhibit a $(k+1)$-chain d with $\partial d = z + z'$. Usually d is found by drawing $z + z'$ and filling it in.

- **Not a boundary.** To show $z \notin B_k$, find a property shared by all boundaries but not by z. When K has few $(k+1)$-simplices one can simply list B_k, which is spanned by the columns of ∂_{k+1}. A typical property: some k-simplex of z is not a face of any $(k+1)$-simplex, so it cannot appear in any boundary.

Example 3.7 (A full computation)**.** Let K be the complex with maximal simplices $\{1,2,3\}$, $\{2,3,4\}$, $\{4,5\}$ and $\{1,5\}$ (Figure 3). It consists of two triangles forming a square $1,2,4,3$, with a path $4-5-1$ attached at two of its corners. Its simplices are the 5 vertices, the 7 edges

$$\{1,2\},\ \{1,3\},\ \{1,5\},\ \{2,3\},\ \{2,4\},\ \{3,4\},\ \{4,5\},$$

and the 2 triangles, so $f = (5,7,2)$ and $\chi = 0$. The boundary matrix ∂_2 is

$$\partial_2 = \begin{array}{c|cc} & 123 & 234 \\ \hline 12 & 1 & 0 \\ 13 & 1 & 0 \\ 15 & 0 & 0 \\ 23 & 1 & 1 \\ 24 & 0 & 1 \\ 34 & 0 & 1 \\ 45 & 0 & 0 \end{array}$$

The 1-skeleton is connected, so $\operatorname{rank}\partial_1 = 5-1 = 4$. The two columns of ∂_2 are nonzero and different, so $\operatorname{rank}\partial_2 = 2$. Hence

$$b_0 = 5-0-4 = 1, \qquad b_1 = 7-4-2 = 1, \qquad b_2 = 2-2-0 = 0,$$

and indeed $b_0 - b_1 + b_2 = 0 = \chi$.

Consider the 1-cycles $z_1 = \{1,2\}+\{2,4\}+\{4,5\}+\{1,5\}$ and $z_2 = \{1,3\}+\{3,4\}+\{4,5\}+\{1,5\}$, which go around the attached path through the two sides of the square. Both are cycles, since every vertex lies on 0 or 2 of their edges. They are homologous:

$$z_1 + z_2 = \{1,2\} + \{2,4\} + \{1,3\} + \{3,4\} = \partial_2\big(\{1,2,3\} + \{2,3,4\}\big).$$

But z_1 is not a boundary: $B_1 = \{0,\ \partial\{1,2,3\},\ \partial\{2,3,4\},\ \partial\{1,2,3\}+\partial\{2,3,4\}\}$, and none of these four chains contains the edge $\{4,5\}$, which lies in no triangle. So $[z_1] = [z_2]$ is the nonzero element of $H_1(K;\mathbb{Z}/2) \cong \mathbb{Z}/2$.

4 Computing Ranks by Row Reduction

Over the field $\mathbb{Z}/2$, ranks are computed by Gaussian elimination, with the simplification that the only nonzero scalar is 1: the elementary operations are swapping two rows and adding one row to another, where addition is entrywise modulo 2 (that is, XOR). The rank is the number of pivots of the resulting echelon form. Column operations may be used as well, since they also preserve the rank.

Example 4.1 (The hollow square)**.** Let K be the 4-cycle with vertices $1,2,3,4$ and edges $\{1,2\},\{2,3\},\{3,4\},\{1,4\}$. Then

$$\partial_1 = \begin{bmatrix} 1&0&0&1 \\ 1&1&0&0 \\ 0&1&1&0 \\ 0&0&1&1 \end{bmatrix} \xrightarrow{R_2+R_1} \begin{bmatrix} 1&0&0&1 \\ 0&1&0&1 \\ 0&1&1&0 \\ 0&0&1&1 \end{bmatrix} \xrightarrow{R_3+R_2} \begin{bmatrix} 1&0&0&1 \\ 0&1&0&1 \\ 0&0&1&1 \\ 0&0&1&1 \end{bmatrix} \xrightarrow{R_4+R_3} \begin{bmatrix} 1&0&0&1 \\ 0&1&0&1 \\ 0&0&1&1 \\ 0&0&0&0 \end{bmatrix}.$$

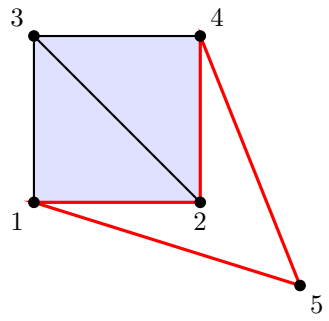


Figure 3: The complex of Example 3.7: two triangles forming a square, and a path $4-5-1$ attached at two corners. The red loop z_1 is a cycle that is not a boundary, and its class spans H_1.

There are three pivots, so $\operatorname{rank}\partial_1 = 3$, $b_0 = 4-3 = 1$ and $b_1 = 4-3-0 = 1$. The kernel of the last matrix is spanned by $(1,1,1,1)^T$, which is the cycle $\{1,2\}+\{2,3\}+\{3,4\}+\{1,4\}$ going once around the square.

If we add the triangles $\{1,2,3\}$ and $\{1,3,4\}$ (and the edge $\{1,3\}$), the square is filled in: ∂_2 has two independent columns, $\operatorname{rank}\partial_1 = 3$ still (the graph is still connected, now with 5 edges), and $b_1 = 5-3-2 = 0$, $b_2 = 2-2 = 0$.

In practice, one rarely needs to reduce ∂_1 by hand, because its rank is given by Theorem 5.1 below: $\operatorname{rank}\partial_1 = n_0 - (\text{number of components})$. For ∂_2, if there are few triangles it is often easiest to decide directly which sets of columns are dependent: a set of columns of ∂_2 sums to zero exactly when the corresponding triangles form a 2-cycle. So $\operatorname{rank}\partial_2 = n_2 - \dim Z_2$, and Z_2 can often be found by the reasoning of Example 3.5.

Remark 4.2. In Week 6 we study a particular column reduction algorithm for boundary matrices, which processes the simplices in a chosen order. Its output contains more than the ranks: it pairs simplices with each other, and this pairing becomes the persistence diagram. Over $\mathbb{Z}$, the analogous computation is the Smith normal form, which also reveals torsion (Section 9).

5 H_0 Counts Connected Components

Theorem 5.1. *For every finite simplicial complex K, b_0 is the number of connected components of $|K|$. Equivalently, if the 1-skeleton of K has c connected components as a graph, then $\operatorname{rank}\partial_1 = n_0 - c$ and $b_0 = c$.*

Proof. Since $\partial_0 = 0$, we have $H_0 = C_0/B_0$, where B_0 is spanned by the boundaries $\{u\}+\{v\}$ of edges $\{u,v\}$. If u and w are joined by a path $u = v_0, v_1, \dots, v_m = w$ in the 1-skeleton, then

$$\{u\}+\{w\} = \partial_1\big(\{v_0,v_1\}+\{v_1,v_2\}+\cdots+\{v_{m-1},v_m\}\big) \in B_0,$$

so $\{u\}$ and $\{w\}$ represent the same class in H_0. Hence H_0 is spanned by one vertex from each component, and $b_0 \le c$. Conversely, consider the linear map $C_0 \to (\mathbb{Z}/2)^c$ that sends a 0-chain to the vector of parities of its numbers of vertices in each component. The boundary of an edge has its two vertices in the same component, so this map vanishes on B_0. It sends the chosen vertex of the i-th component to the i-th standard basis vector, so the c chosen vertices are independent in H_0, and $b_0 \ge c$. Finally, the components of $|K|$ are the same as those of the 1-skeleton, since every simplex is connected and contains its edges. □

In particular, b_0 can be computed by a graph search or by the union-find algorithm, without any linear algebra. The proof also shows a fact we will use: *the boundary of any* 1*-chain contains an even number of vertices in each component.* In particular, a single vertex is never a boundary.

6 The Euler–Poincaré Formula

Theorem 6.1 (Euler–Poincaré)**.** *For every finite simplicial complex K of dimension d,*

$$\chi(K) = \sum_{k=0}^{d} (-1)^k n_k = \sum_{k=0}^{d} (-1)^k b_k.$$

Proof. Write $r_k = \operatorname{rank} \partial_k$, with $r_0 = 0$ (since $\partial_0 = 0$) and $r_{d+1} = 0$ (since $C_{d+1} = 0$). By rank–nullity applied to $\partial_k : C_k \to C_{k-1}$, $n_k = \dim Z_k + r_k$, and since $B_k \subseteq Z_k$, $b_k = \dim Z_k - r_{k+1}$. Hence $n_k - b_k = r_k + r_{k+1}$ and

$$\sum_{k=0}^{d} (-1)^k (n_k - b_k) = \sum_{k=0}^{d} (-1)^k (r_k + r_{k+1}) = r_0 + (-1)^d r_{d+1} = 0,$$

because each r_k with $1 \le k \le d$ appears twice, with signs $(-1)^k$ and $(-1)^{k-1}$, and cancels. □

The individual counts n_k depend on the triangulation, but the Betti numbers do not (Theorem 10.1). The Euler–Poincaré formula therefore proves that the Euler characteristic is a topological invariant. It is also a computational shortcut: if all Betti numbers but one are known, the remaining one follows from χ, which only requires counting simplices.

Proposition 6.2. *Let K be a triangulation of a closed connected surface. Then $b_0 = b_2 = 1$ over $\mathbb{Z}/2$, and hence $b_1 = 2 - \chi(K)$.*

Proof. $b_0 = 1$ by Theorem 5.1. There are no 3-simplices, so $b_2 = \dim Z_2$. A 2-chain is a set S of triangles, and $\partial_2 S = 0$ means that every edge lies in an even number of triangles of S. Since every edge of K lies in exactly two triangles, this says: if a triangle is in S, then so is its neighbor across each of its edges. In a connected closed surface any two triangles can be joined by a chain of triangles, consecutive ones sharing an edge. So either $S = \emptyset$ or S consists of all triangles, and both are cycles. Hence $\dim Z_2 = 1$. The last claim follows from Theorem 6.1. □

By the classification of surfaces, over $\mathbb{Z}/2$ we obtain $b_1 = 0$ for the sphere, $b_1 = 2g$ for the orientable surface of genus g, and $b_1 = k$ for the connected sum of k projective planes. In particular the torus and the Klein bottle both have Betti numbers $(1, 2, 1)$ over $\mathbb{Z}/2$, and $\mathbb{R}P^2$ has $(1, 1, 1)$. *Over $\mathbb{Z}/2$, homology cannot see orientability*; for that we need integer coefficients.

Example 6.3 (Counting)**.** For a triangulation of a closed surface, each triangle has three edges and each edge lies in exactly two triangles, so $3n_2 = 2n_1$. Hence the f-vector is determined by n_0 and n_1, and so is $\chi = n_0 - n_1/3$. For example:

- A closed connected surface triangulated with 8 vertices and 24 edges has $n_2 = 16$ and $\chi = 0$, so it is the torus or the Klein bottle, with $b = (1, 2, 1)$ over $\mathbb{Z}/2$.
- If every vertex of a triangulated closed surface lies on exactly d edges, then $2n_1 = d\, n_0$, so $\chi = n_0(1 - d/2 + d/3) = n_0(6 - d)/6$. For $d = 5$ this is $n_0/6 > 0$, which forces the sphere and $n_0 = 12$ (the icosahedron), or $\mathbb{R}P^2$ and $n_0 = 6$.

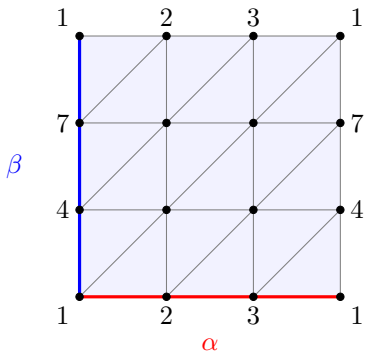


Figure 4: Two 1-cycles on the 9-vertex torus: $\alpha = \{1,2\} + \{2,3\} + \{3,1\}$ along the bottom edge of the square and $\beta = \{1,4\} + \{4,7\} + \{7,1\}$ along the left edge. Their classes form a basis of $H_1(T^2; \mathbb{Z}/2)$, which has dimension 2. The sum of all 18 triangles is a 2-cycle, which spans H_2.

7 Building Blocks: Unions, One-Point Unions and Cones

Many complexes are assembled from simpler pieces, and their homology can then be computed piece by piece.

Proposition 7.1 (Disjoint unions)**.** *If $K = K_1 \sqcup K_2$ is a disjoint union of subcomplexes, then $b_k(K) = b_k(K_1) + b_k(K_2)$ for every k.*

Proof. Every simplex lies in exactly one of K_1, K_2, so $C_k(K) = C_k(K_1) \oplus C_k(K_2)$, and ∂ maps each summand into the corresponding summand. Hence the boundary matrices are block diagonal, and ranks add. □

Proposition 7.2 (One-point unions)**.** *Let $K = K_1 \cup K_2$ where K_1, K_2 are subcomplexes with $K_1 \cap K_2 = \{\{v\}\}$, a single vertex. Then*

$$b_0(K) = b_0(K_1) + b_0(K_2) - 1, \qquad b_k(K) = b_k(K_1) + b_k(K_2) \quad \text{for } k \geq 1.$$

Proof. For $k \geq 1$, no k-simplex lies in both K_1 and K_2, so $C_k(K) = C_k(K_1) \oplus C_k(K_2)$, and ∂_k maps $C_k(K_i)$ into $C_{k-1}(K_i)$. For $k \geq 2$ the boundary matrices ∂_k are therefore block diagonal, so B_k for $k \geq 1$ and Z_k for $k \geq 2$ split as direct sums.

It remains to see that Z_1 splits. Let $c = c_1 + c_2$ be a 1-cycle with $c_i \in C_1(K_i)$. Then $\partial c_1 = \partial c_2$ (we are over $\mathbb{Z}/2$), and this 0-chain lies in $C_0(K_1) \cap C_0(K_2)$, so it is 0 or $\{v\}$. A boundary of a 1-chain contains an even number of vertices, so $\partial c_1 = \partial c_2 = 0$, that is, $c_i \in Z_1(K_i)$. Thus $Z_1(K) = Z_1(K_1) \oplus Z_1(K_2)$, and $H_k(K) \cong H_k(K_1) \oplus H_k(K_2)$ for all $k \geq 1$. The formula for b_0 holds because the components of K_1 and K_2 containing v merge into one. □

Example 7.3.

- Two hollow triangles sharing a vertex (a "figure eight"): $b = (1, 1+1) = (1, 2)$. Check: $f = (5, 6)$, $\chi = -1 = 1 - 2$.
- Let K be the boundary of the tetrahedron on $\{0,1,2,3\}$ together with the hollow triangle on $\{0,4,5\}$, glued at the vertex 0: a sphere with a circle attached. Then $b = (1, 0+1, 1+0) = (1,1,1)$. Check: $f = (6, 9, 4)$ and $\chi = 6 - 9 + 4 = 1 = 1 - 1 + 1$.

Definition 7.4. A simplicial complex K is a **cone** with apex v if $\sigma \cup \{v\} \in K$ for every $\sigma \in K$.

Examples of cones: a full simplex (any vertex is an apex); the closed star $\overline{\mathrm{St}}(v)$ of a vertex v, with apex v; the complex obtained from any complex L by adding a new vertex v and all simplices $\sigma \cup \{v\}$, $\sigma \in L$ (the *cone over* L). Geometrically a cone can be contracted to its apex, so it should have the homology of a point. Here is a direct algebraic proof.

Proposition 7.5 (Cones are acyclic)**.** *If K is a cone with apex v, then $b_0(K) = 1$ and $b_k(K) = 0$ for all $k \geq 1$.*

Proof. Every vertex w is joined to v by the edge $\{v, w\}$, so K is connected and $b_0 = 1$. For a simplex σ not containing v, write $v * \sigma = \sigma \cup \{v\}$, and extend linearly to chains not involving v. For a j-simplex σ with $j \geq 1$ not containing v, the faces of $v * \sigma$ are σ itself and the simplices $v * \tau$ with τ a facet of σ, so

$$\partial(v * \sigma) = \sigma + v * \partial\sigma,$$

and by linearity $\partial(v * a) = a + v * \partial a$ for every chain a of dimension ≥ 1 not involving v.

Let $k \geq 1$ and let z be a k-cycle. Split $z = a + v * b$, where a consists of the simplices of z not containing v, and b is a $(k-1)$-chain not involving v. Look at the part of $\partial z = 0$ that does not involve v: it is $\partial a + b$, because $\partial(v * b) = b +$ (terms containing v). So $\partial a = b$, and therefore

$$\partial(v * a) = a + v * \partial a = a + v * b = z.$$

Thus every k-cycle is a boundary, $H_k(K; \mathbb{Z}/2) = 0$. □

Corollary 7.6. *1. The full simplex on any finite vertex set has $b_0 = 1$ and $b_k = 0$ for $k \geq 1$.*

2. If X is a finite set in which all pairwise distances are at most 2ϵ, then $VR_\epsilon(X)$ is the full simplex on X, and so has the homology of a point.

3. If some point $x_0 \in X$ satisfies $d(x_0, x) \leq 2\epsilon$ for every $x \in X$, then $VR_\epsilon(X)$ is a cone with apex x_0 (since VR is a flag complex), and again has the homology of a point.

8 Homology of Čech and Vietoris–Rips Complexes

For a point cloud X, the complexes $\check{C}_\epsilon(X)$ and $VR_\epsilon(X)$ only change at finitely many values of ϵ: the edge $\{x, y\}$ enters VR_ϵ at $\epsilon = d(x, y)/2$, and a simplex enters $\check{C}_\epsilon$ when ϵ reaches the radius of its smallest enclosing ball. To describe these complexes "for every $\epsilon > 0$" one lists the critical values, describes the complex on each interval between them, and computes its Betti numbers. Holes that exist on a long interval of ϵ are features of the data; this is the idea of persistent homology.

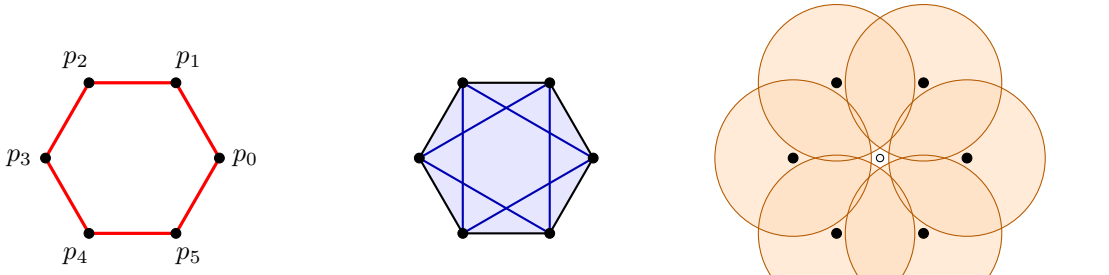
Example 8.1 (The regular hexagon)**.** Let $X = \{p_0, \ldots, p_5\}$ be the vertices of a regular hexagon with side 1, labelled cyclically. The pairwise distances are 1 (neighbors), $\sqrt{3}$ (second neighbors) and 2 (opposite vertices). Hence the Vietoris–Rips complex is as follows.

- $0 < \epsilon < \frac{1}{2}$: six points, $b = (6)$.
- $\frac{1}{2} \leq \epsilon < \frac{\sqrt{3}}{2}$: the six sides, a hexagon; there are no triangles, since every triangle contains a pair at distance $\geq \sqrt{3}$. So $b_0 = 1$, $b_1 = 1$.

- $\frac{\sqrt{3}}{2} \le \epsilon < 1$: every pair except the three opposite pairs $\{p_0, p_3\}$, $\{p_1, p_4\}$, $\{p_2, p_5\}$ is an edge, so $n_1 = 15-3 = 12$. A triple is a simplex if and only if it contains no opposite pair, i.e. it picks one vertex from each opposite pair: $2^3 = 8$ triangles. No 4 vertices avoid all opposite pairs, so there are no tetrahedra. This is exactly the boundary of the octahedron, with opposite pairs as the antipodal vertices, so $f = (6, 12, 8)$ and $b = (1, 0, 1)$.
- $\epsilon \ge 1$: all pairs, the full 5-simplex, $b = (1, 0, 0, \dots)$ by Corollary 7.6.

Now compare with the Čech complex for $\frac{\sqrt{3}}{2} \le \epsilon < 1$. Its edges are the same as those of VR_ϵ (a pair of points has enclosing radius half its distance). A triple of consecutive vertices $\{p_i, p_{i+1}, p_{i+2}\}$ forms an obtuse triangle with longest side $\sqrt{3}$, so its smallest enclosing disk has radius $\frac{\sqrt{3}}{2} \le \epsilon$ and it is in $\check{C}_\epsilon$. The triples $\{p_0, p_2, p_4\}$ and $\{p_1, p_3, p_5\}$ form equilateral triangles of side $\sqrt{3}$, with circumradius $1 > \epsilon$, so they are *not* in $\check{C}_\epsilon$. Hence $\check{C}_\epsilon$ has $f = (6, 12, 6)$, $\chi = 0$, and one computes $b = (1, 1, 0)$. This agrees with the Nerve Theorem: the union of the six disks of radius $\epsilon < 1$ misses the center of the hexagon, which is at distance 1 from all six points, so the union is an annulus.

So in this range VR_ϵ has a 2-dimensional void that does not exist in the data, while the Čech complex correctly sees a 1-dimensional hole. This is consistent with Proposition 1.1: $VR_\epsilon \subseteq \check{C}_{\sqrt{2}\epsilon}$, and $\sqrt{2}\epsilon \ge \frac{\sqrt{6}}{2} > 1$, where the Čech complex is already a full simplex.



VR_ϵ, $\frac{1}{2} \le \epsilon < \frac{\sqrt{3}}{2}$: a hexagon, $b = (1, 1)$

VR_ϵ, $\frac{\sqrt{3}}{2} \le \epsilon < 1$: an octahedron, $b = (1, 0, 1)$

disks of radius $\epsilon < 1$: the center is uncovered; $\check{C}_\epsilon$ has $b = (1, 1, 0)$

Figure 5: The regular hexagon of Example 8.1. In the middle panel the two blue triangles are edges of length $\sqrt{3}$; together with the sides they form the 1-skeleton of an octahedron, and VR_ϵ is its boundary, a 2-sphere. The union of disks (right) is an annulus, and the Čech complex has its homology.

Example 8.2 (Adding a center)**.** Add the center p_c of the hexagon to X. It is at distance 1 from each p_i, so for $\epsilon \ge \frac{1}{2}$ it is joined to all six vertices, at the same moment as the sides appear, and the six triangles $\{p_c, p_i, p_{i+1}\}$ (equilateral of side 1) fill the hexagon. In fact for every $\epsilon \ge \frac{1}{2}$, $VR_\epsilon(X \cup \{p_c\})$ is a cone with apex p_c, by Corollary 7.6(3), so it has the homology of a point; for $\epsilon < \frac{1}{2}$ it is 7 points. The hole of the hexagon is never born: a point in the middle of a hole destroys it. Compare this with Example 3.7, where a hole is killed because the loop around it becomes the boundary of a sum of triangles.

9 Orientations and Integer Coefficients

Not on Midterm 1. On the midterm all homology is over $\mathbb{Z}/2$. From this section only the following are needed, and they are Week 2 material: finding the vertex classes of a polygon word, computing $\chi = V - E + F$, deciding orientability, and identifying the surface by the classification theorem (as in the first paragraph of Example 9.7). The $\mathbb{Z}/2$-Betti numbers of a closed surface then follow from Proposition 6.2.

Over the integers, $1+1 \neq 0$, and the boundary of a path must be its end point *minus* its starting point. This requires keeping track of directions.

Definition 9.1. An **orientation** of a k-simplex, $k \geq 1$, is an ordering of its vertices, where two orderings are considered equivalent if they differ by an even permutation. Each simplex of dimension $k \geq 1$ has exactly two orientations. We write $[v_0, v_1, \ldots, v_k]$ for the oriented simplex, and $-[v_0, \ldots, v_k]$ for the same simplex with the opposite orientation. Thus $[v_0, v_1] = -[v_1, v_0]$ and $[v_0, v_1, v_2] = [v_1, v_2, v_0] = -[v_1, v_0, v_2]$.

In computations one fixes an orientation of every simplex, usually the one given by increasing vertex labels.

Definition 9.2. The group of **k-chains with integer coefficients** $C_k(K; \mathbb{Z})$ consists of the formal sums $\sum a_\sigma \sigma$ with $a_\sigma \in \mathbb{Z}$, one term for each k-simplex with its chosen orientation. The **boundary operator** is

$$\partial_k [v_0, \ldots, v_k] = \sum_{i=0}^{k} (-1)^i [v_0, \ldots, \widehat{v_i}, \ldots, v_k],$$

extended linearly.

Interchanging two vertices changes the sign of the right-hand side, so ∂_k is well defined on oriented simplices. For example,

$$\partial_1[v_0, v_1] = v_1 - v_0, \qquad \partial_2[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1] = [v_0, v_1] + [v_1, v_2] + [v_2, v_0].$$

The boundary of an oriented triangle is the loop around it, traversed in the direction of the orientation.

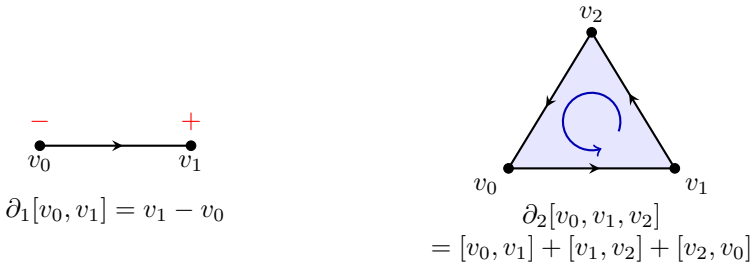


Figure 6: Oriented simplices. An orientation of an edge is a direction, and its boundary is the end point minus the starting point. An orientation of a triangle is a sense of rotation, and its boundary is the loop around it in that direction.

Lemma 9.3. *Over* $\mathbb{Z}$, $\partial_{k-1} \circ \partial_k = 0$.

Proof. Let $\sigma = [v_0, \dots, v_k]$ and $i < j$. The face obtained by deleting v_i and v_j appears twice in $\partial\partial\sigma$. Deleting v_i first gives the sign $(-1)^i$, and then v_j, now in position $j-1$, gives $(-1)^{j-1}$. Deleting v_j first gives $(-1)^j$, and then v_i, still in position i, gives $(-1)^i$. The total signs $(-1)^{i+j-1}$ and $(-1)^{i+j}$ are opposite, so the two terms cancel. □

We define cycles $Z_k = \ker \partial_k$, boundaries $B_k = \operatorname{im} \partial_{k+1}$, and the **integral homology groups** $H_k(K;\mathbb{Z}) = Z_k/B_k$. These are finitely generated abelian groups, so

$$H_k(K;\mathbb{Z}) \cong \mathbb{Z}^{\beta_k} \oplus \mathbb{Z}/m_1 \oplus \cdots \oplus \mathbb{Z}/m_s$$

for an integer $\beta_k \geq 0$, the **rank** of H_k, and integers $m_i \geq 2$; the finite part is the **torsion** of H_k. Integral homology is computed from the boundary matrices by the Smith normal form.

Example 9.4 (The circle over $\mathbb{Z}$)**.** For the hollow triangle, orient the edges as $[1,2]$, $[2,3]$, $[1,3]$. A 1-chain $a[1,2] + b[2,3] + c[1,3]$ has boundary

$$a(2-1) + b(3-2) + c(3-1) = -(a+c)\,1 + (a-b)\,2 + (b+c)\,3,$$

which vanishes exactly when $b = a$ and $c = -a$. So $Z_1 = \mathbb{Z} \cdot z$ with $z = [1,2] + [2,3] + [3,1]$, the loop traversed once, and $H_1 \cong \mathbb{Z}$.

	over $\mathbb{Z}$			over $\mathbb{Z}/2$		
Space	H_0	H_1	H_2	b_0	b_1	b_2
point, disk, cone	$\mathbb{Z}$	0	0	1	0	0
circle	$\mathbb{Z}$	$\mathbb{Z}$	0	1	1	0
sphere S^2	$\mathbb{Z}$	0	$\mathbb{Z}$	1	0	1
torus T^2	$\mathbb{Z}$	$\mathbb{Z}^2$	$\mathbb{Z}$	1	2	1
orientable surface of genus g	$\mathbb{Z}$	$\mathbb{Z}^{2g}$	$\mathbb{Z}$	1	$2g$	1
projective plane $\mathbb{R}P^2$	$\mathbb{Z}$	$\mathbb{Z}/2$	0	1	1	1
Klein bottle	$\mathbb{Z}$	$\mathbb{Z} \oplus \mathbb{Z}/2$	0	1	2	1
$\#^k \mathbb{R}P^2$	$\mathbb{Z}$	$\mathbb{Z}^{k-1} \oplus \mathbb{Z}/2$	0	1	k	1
Möbius band	$\mathbb{Z}$	$\mathbb{Z}$	0	1	1	0

- **Integer homology distinguishes the torus from the Klein bottle**, while $\mathbb{Z}/2$-homology does not.
- H_2 **detects orientability.** For a closed connected surface, $H_2(K;\mathbb{Z}) \cong \mathbb{Z}$ if the surface is orientable and $H_2(K;\mathbb{Z}) = 0$ otherwise. Indeed, an integral 2-cycle must cancel along every edge, so its coefficients on two neighboring triangles are equal up to a sign determined by whether their chosen orientations agree along the common edge. Propagating through the surface, a nonzero 2-cycle exists if and only if the triangles can be oriented consistently, and then the sum of all consistently oriented triangles generates H_2.
- **Torsion comes from twisting.** In $\mathbb{R}P^2$, presented by the word aa, the loop a is not a boundary, but the boundary of the disk traverses it twice, so $2a$ is a boundary and $H_1 = \mathbb{Z}/2$. Over $\mathbb{Z}/2$, where $2a = 0$ anyway, this relation is invisible, and the torsion reappears as an extra class in H_1 and in H_2. This is why b_k over $\mathbb{Z}/2$ can be larger than the rank β_k over $\mathbb{Z}$: for $\#^k \mathbb{R}P^2$, $b_1 = k$ while $\beta_1 = k - 1$.

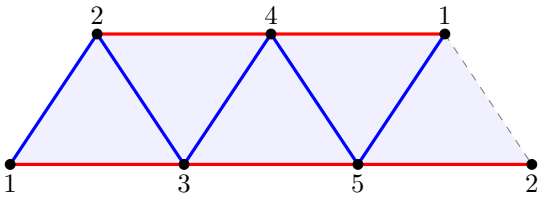


Figure 7: The 5-vertex Möbius band; the dashed end edge is glued to the edge $\{1, 2\}$ on the left with a half-twist. The core loop $1-2-3-4-5-1$ (blue) runs once along the band; the boundary loop $1-3-5-2-4-1$ (red) goes around twice. In $H_1(\,\cdot\,;\mathbb{Z})$ the boundary loop equals twice the core loop.

9.1 First homology of surfaces from polygon words

For surfaces given by polygon presentations, H_1 can be read off from the boundary word. We state the result without proof; it follows from cellular homology, which computes homology from a cell decomposition with fewer cells than a triangulation.

Proposition 9.5. *Suppose a closed surface is given by a polygon with boundary word w in which* all corners are identified to a single vertex. *Then $H_1(\,\cdot\,;\mathbb{Z})$ is the abelian group generated by the letters of w, subject to the single relation obtained by adding up the letters of w with their exponents as signs.*

Example 9.6.

- Torus, $w = aba^{-1}b^{-1}$: the relation $a + b - a - b = 0$ is trivial, so $H_1 \cong \mathbb{Z}^2$.
- Projective plane, $w = aa$: the relation $2a = 0$ gives $H_1 \cong \mathbb{Z}/2$.
- Klein bottle, $w = aba^{-1}b$: the relation $2b = 0$ gives $H_1 \cong \mathbb{Z} \oplus \mathbb{Z}/2$.
- Genus g, $w = a_1b_1a_1^{-1}b_1^{-1} \cdots a_gb_ga_g^{-1}b_g^{-1}$: $H_1 \cong \mathbb{Z}^{2g}$.
- k projective planes, $w = a_1a_1 \cdots a_ka_k$: the relation $2(a_1 + \cdots + a_k) = 0$. With the new basis $a_1, \ldots, a_{k-1}$, $s = a_1 + \cdots + a_k$ the relation reads $2s = 0$, so $H_1 \cong \mathbb{Z}^{k-1} \oplus \mathbb{Z}/2$.

Example 9.7 (A worked example)**.** Let M be given by the hexagon with boundary word $w = a\,a\,b^{-1}c\,b\,c$. Label the corners $P_0, \ldots, P_5$ so that the sides are $P_0P_1 = a$, $P_1P_2 = a$, $P_2P_3 = b^{-1}$ (so b runs from P_3 to P_2), $P_3P_4 = c$, $P_4P_5 = b$, $P_5P_0 = c$. The identifications of corners are

$$a : P_0 \sim P_1,\ P_1 \sim P_2; \qquad b : P_3 \sim P_4,\ P_2 \sim P_5; \qquad c : P_3 \sim P_5,\ P_4 \sim P_0,$$

and chaining them gives a single vertex class. So $V = 1$, $E = 3$, $F = 1$, and $\chi(M) = -1$. The letter a occurs twice with the same exponent, so M is non-orientable, and by the classification $M \cong \#^3\mathbb{R}P^2$.

By Proposition 9.5, $H_1(M;\mathbb{Z})$ is generated by a, b, c with the relation $2a-b+c+b+c = 2a+2c = 0$. In the basis a, b, $s = a + c$ of $\mathbb{Z}^3$ the relation is $2s = 0$, so

$$H_1(M;\mathbb{Z}) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/2,$$

with free part generated by a, b and torsion generated by $a + c$. Over $\mathbb{Z}/2$, $b_0 = b_2 = 1$ by Proposition 6.2, and $b_1 = 2 - \chi = 3$; directly, the relation $2a + 2c$ vanishes mod 2, so $H_1(M;\mathbb{Z}/2) \cong (\mathbb{Z}/2)^3$. Since M is non-orientable, $H_2(M;\mathbb{Z}) = 0$.

Remark 9.8 (Check the vertices first). The hypothesis of Proposition 9.5 is essential. For the square with word $abab$, the corners fall into *two* classes $\{P_0, P_2\}$ and $\{P_1, P_3\}$, so $\chi = 2 - 2 + 1 = 1$ and the surface is $\mathbb{R}P^2$ (indeed $abab = (ab)(ab)$, which is cc with $c = ab$). Its first homology is $\mathbb{Z}/2$, not the group $\mathbb{Z} \oplus \mathbb{Z}/2$ that the naive recipe "generators a, b, relation $2a + 2b = 0$" would give. With more than one vertex, the edges between different vertices have nonzero boundary and the recipe no longer applies; first transform the word into one with a single vertex.

10 Invariance and Functoriality

Not on Midterm 1. You may use, without proof, that Betti numbers do not depend on the triangulation (Theorem 10.1), as in Proposition 6.2. Functoriality and the Brouwer theorem are not on the midterm.

The homology groups are defined using a triangulation, but they do not depend on it. We state this fundamental theorem without proof; the proof uses simplicial approximation (see Munkres, *Elements of Algebraic Topology*).

Theorem 10.1 (Topological invariance)**.** *If $|K|$ and $|L|$ are homeomorphic, then $H_k(K;\mathbb{Z}) \cong H_k(L;\mathbb{Z})$ and $H_k(K;\mathbb{Z}/2) \cong H_k(L;\mathbb{Z}/2)$ for all k. More generally, the same holds if $|K|$ and $|L|$ are only homotopy equivalent.*

For instance, the boundaries of the tetrahedron and of the octahedron have the same homology, $b = (1, 0, 1)$. Homotopy invariance gives another proof that a cone has the homology of a point: it is contractible. Combined with the Nerve Theorem, it says that $\check{C}_\epsilon(X)$ has the homology of the union of balls of radius ϵ.

A **simplicial map** $f : K \to L$ is a map on vertices that sends simplices to simplices (possibly of lower dimension). It induces $f_\# : C_k(K) \to C_k(L)$, sending σ to $f(\sigma)$ if $f(\sigma)$ is still a k-simplex and to 0 otherwise. One checks that $\partial \circ f_\# = f_\# \circ \partial$, so $f_\#$ sends cycles to cycles and boundaries to boundaries, and induces

$$f_* : H_k(K) \to H_k(L), \qquad f_*[z] = [f_\# z],$$

with $(g \circ f)_* = g_* \circ f_*$ and $(\mathrm{id})_* = \mathrm{id}$. This is **functoriality**. Using simplicial approximation, every continuous map $|K| \to |L|$ also induces maps in homology with the same properties. The inclusion of a subcomplex $L \subseteq K$ is a simplicial map; the induced map $H_k(L) \to H_k(K)$ is the one used in persistent homology to follow holes as ϵ grows.

Functoriality is a powerful tool for proving that certain maps cannot exist. Recall that $H_n(S^n;\mathbb{Z}) \cong \mathbb{Z}$ for $n \geq 1$ (the boundary of the $(n+1)$-simplex), while the ball D^n (a full simplex) has the homology of a point.

Theorem 10.2 (No retraction; Brouwer)**.** *For $n \geq 1$ there is no continuous map $r : D^n \to S^{n-1}$ with $r(x) = x$ for all $x \in S^{n-1}$. Consequently, every continuous map $f : D^n \to D^n$ has a fixed point.*

Proof. Suppose such a retraction r exists and let $i : S^{n-1} \to D^n$ be the inclusion. Then $r \circ i = \mathrm{id}_{S^{n-1}}$, so by functoriality $r_* \circ i_* = \mathrm{id}$ on $H_{n-1}(S^{n-1};\mathbb{Z}) \cong \mathbb{Z}$ (for $n \geq 2$). But i_* factors through $H_{n-1}(D^n;\mathbb{Z}) = 0$, so $r_* \circ i_* = 0$, a contradiction. For $n = 1$, use H_0 instead: S^0 is two points, $b_0 = 2$, while D^1 is connected, and a continuous map cannot send a connected space onto two points.

If $f : D^n \to D^n$ has no fixed point, then for each x the ray from $f(x)$ through x meets S^{n-1} in a unique point $r(x)$ beyond x. The map r is continuous and $r(x) = x$ for $x \in S^{n-1}$, which is impossible. □

11 Local Homology and Links

Not on Midterm 1. Computing the link of a vertex (Section 1.2) is on the midterm; using links to decide whether a space is a surface is not.

Homology can also detect what a space looks like near a point. We state the key fact without proof. For a point x of $|K|$, subdivide K if necessary so that x is a vertex. Then the homology of the link $\mathrm{lk}(x)$ does not depend on the triangulation, and it depends only on a neighborhood of x up to homeomorphism. (It is the **local homology** of $|K|$ at x, shifted by one degree.) The reference cases are:

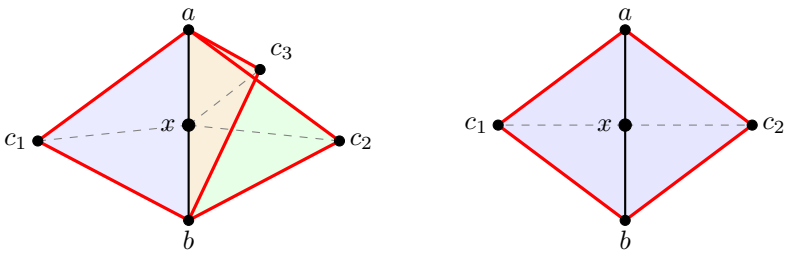
- a point of an open disk (an interior point of a surface): the link is a circle, $b = (1, 1)$;
- a boundary point of a surface: the link is a path, $b = (1, 0)$;
- a point in the interior of an n-dimensional ball: the link is an $(n-1)$-sphere.

Hence: *if x is a vertex of K whose link has Betti numbers different from $(1, 1)$ and $(1, 0)$, then no neighborhood of x is homeomorphic to an open disk or a half-disk, and $|K|$ is not a surface.*

Example 11.1 (The edge lemma)**.** Let x be a point in the interior of an edge $\{a, b\}$ that lies in exactly $k \geq 1$ triangles $\{a, b, c_1\}, \ldots, \{a, b, c_k\}$. Subdivide the edge at x, which splits each of these triangles into two. The link of x is then the graph with the $k + 2$ vertices $a, b, c_1, \ldots, c_k$ and the $2k$ edges $\{a, c_i\}$, $\{b, c_i\}$. It is connected, so by Example 3.6

$$b_1(\mathrm{lk}(x)) = 2k - (k + 2) + 1 = k - 1.$$

So x has a disk neighborhood only if $k = 2$, and a half-disk neighborhood only if $k = 1$. This proves that in a triangulated closed surface every edge lies in exactly two triangles.



$k = 3$ triangles on an edge: link of x has $b_1 = 2$

$k = 2$ triangles on an edge: link of x is a circle, $b_1 = 1$

Figure 8: Local homology at a point x in the interior of an edge $\{a, b\}$. After subdividing at x (dashed edges), the link of x (red) is a graph with $k + 2$ vertices and $2k$ edges, so $b_1 = k - 1$. Only $k = 2$ gives the circle of an interior point of a surface.

Example 11.2 (The bowtie)**.** Let K consist of the two triangles $\{0,1,2\}$ and $\{0,3,4\}$ and their faces, sharing only the vertex 0. By Proposition 7.2 and Proposition 7.5 it has the homology of a point. Its vertex 0 has

$$\mathrm{lk}(0) = \big\{\{1\},\{2\},\{3\},\{4\},\{1,2\},\{3,4\}\big\},$$

two disjoint edges, with $b = (2,0)$. This is neither $(1,1)$ nor $(1,0)$, so $|K|$ is not a surface, even allowing boundary: the vertex 0 is a "pinch point". Note that homology of K itself cannot see this, since K has the homology of a point, exactly like a disk; it is the link that detects the singularity.

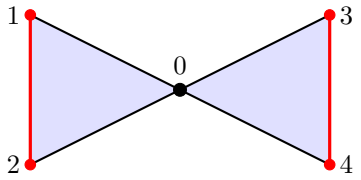


Figure 9: The bowtie of Example 11.2. The link of the vertex 0 (red) consists of two disjoint edges, so 0 has no disk or half-disk neighborhood.

12 Summary: Computational Recipes

1. **From maximal simplices to the f-vector.** List all faces of the maximal simplices without repetition; count by dimension; $\chi = \sum(-1)^k n_k$.

2. **Betti numbers over $\mathbb{Z}/2$.** $b_k = n_k - \operatorname{rank}\partial_k - \operatorname{rank}\partial_{k+1}$. Use $\operatorname{rank}\partial_1 = n_0 - (\#\text{components})$. For ∂_2, find the 2-cycles: $\operatorname{rank}\partial_2 = n_2 - \dim Z_2$.

3. **Shortcuts.** b_0 = number of components. If $\dim K = d$, then $b_d = \dim Z_d$. One unknown Betti number follows from Euler–Poincaré. For a closed connected surface $b_0 = b_2 = 1$ and $b_1 = 2 - \chi$; for a connected graph $b_1 = E - V + 1$.

4. **Building blocks.** Betti numbers add over disjoint unions, and add in degrees ≥ 1 over one-point unions. Cones and full simplices have the homology of a point.

5. **Homologous cycles.** $[z] = [z']$ iff $z + z' = \partial d$; find d by filling in $z + z'$. A cycle containing a k-simplex that is a face of no $(k+1)$-simplex is not a boundary.

6. **Rips and Čech complexes.** Find the critical values ($d(x,y)/2$ for VR, enclosing radii for $\check{C}$), describe the complex between consecutive critical values, then use items 2–4. $\check{C}_\epsilon \subseteq VR_\epsilon \subseteq \check{C}_{\sqrt{2}\epsilon}$ in the plane.

7. **Surfaces from words.** Check that all corners form one vertex class; then **[Not on Midterm 1]** $H_1(\,\cdot\,;\mathbb{Z})$ = free abelian group on the letters modulo the sum of the word; change basis to read off torsion. (On the midterm:) $\chi = V - E + F$, orientability from repeated letters with equal exponents, and $b_0 = b_2 = 1$, $b_1 = 2 - \chi$ over $\mathbb{Z}/2$.

8. **[Not on Midterm 1] Is it a surface?** Compute the link of a suspicious vertex: it must be a circle (interior point) or a path (boundary point).

13 Exercises

Answers are given in brackets.

1. Compute the Betti numbers over $\mathbb{Z}/2$ of: (a) the 1-skeleton of the tetrahedron; (b) the boundary of the octahedron; (c) the annulus with triangles $\{1,2,4\},\{2,4,5\},\{2,3,5\}$, $\{3,5,6\},\{1,3,6\},\{1,4,6\}$.
   [(a) $(1,3)$; (b) $(1,0,1)$; (c) $(1,1,0)$.]

2. Write out all twelve terms of $\partial_2\partial_3\{v_0,v_1,v_2,v_3\}$ over $\mathbb{Z}/2$ and check that they cancel in pairs. **[Not on Midterm 1]** Do the same over $\mathbb{Z}$ for $[v_0,v_1,v_2,v_3]$.

3. Let K have maximal simplices $\{1,2,3\}$, $\{3,4,5\}$ and $\{1,5\}$. List its simplices, write ∂_1 and ∂_2 over $\mathbb{Z}/2$, and compute the Betti numbers. Show that the cycles $\{1,3\}+\{3,5\}+\{1,5\}$ and $\{1,2\}+\{2,3\}+\{3,4\}+\{4,5\}+\{1,5\}$ are homologous and generate H_1.
   [$f=(5,7,2)$, $\chi=0$, $b=(1,1,0)$.]

4. A closed connected surface is triangulated with 12 vertices and 30 edges. Which surface is it? What if it has 10 vertices and 30 edges?
   [$n_2=20$, $\chi=2$: the sphere. $n_2=20$, $\chi=0$: torus or Klein bottle, $b=(1,2,1)$.]

5. Let X consist of the six vertices of a regular hexagon of side 1 and its center. Describe $VR_\epsilon(X)$ for every $\epsilon>0$ and give its Betti numbers. Do the same for $\check{C}_\epsilon(X)$.
   [7 points for $\epsilon<\frac{1}{2}$; a cone with apex the center, hence $b=(1,0,\dots)$, for $\epsilon\geq\frac{1}{2}$. For $\check{C}_\epsilon$: 7 points for $\epsilon<\frac{1}{2}$; for $\frac{1}{2}\leq\epsilon<\frac{1}{\sqrt{3}}$ only the 6 sides and 6 spokes, since a triangle $\{p_c,p_i,p_{i+1}\}$ is equilateral of side 1 with circumradius $\frac{1}{\sqrt{3}}$, so $b=(1,12-7+1)=(1,6)$; for $\epsilon\geq\frac{1}{\sqrt{3}}$ the homology of a point (the union of disks is contractible). So here the two complexes differ even in b_1.]

6. Let $X=\{(0,0),(4,0),(0,3)\}$. Find the values of ϵ at which $VR_\epsilon(X)$ and $\check{C}_\epsilon(X)$ become a full triangle.
   [VR: $\epsilon=\frac{5}{2}$ (longest side 5); $\check{C}$: also $\frac{5}{2}$, since the triangle is right-angled and its smallest enclosing disk has the hypotenuse as diameter.]

7. Show that the closed star of a vertex is a cone, and conclude that it has the homology of a point. Is the same true for the link?

8. Let L be a complex and L' the complex obtained by attaching a hollow triangle to L at one vertex. How do the Betti numbers change? What if a solid triangle is attached instead?
   [b_1 increases by 1; nothing changes.]

9. **[Not on Midterm 1]** Using boundary words, compute $H_1(\,\cdot\,;\mathbb{Z})$ and the $\mathbb{Z}/2$-Betti numbers of: (a) $\#^4\mathbb{R}P^2$, with word $a_1a_1a_2a_2a_3a_3a_4a_4$; (b) the surface with hexagonal word $a\,b\,a^{-1}b^{-1}c\,c$ (first check the vertices). What does (b) say about $T^2\#\mathbb{R}P^2$?
   [(a) $\mathbb{Z}^3\oplus\mathbb{Z}/2$, $(1,4,1)$. (b) One vertex, $\chi=-1$, $H_1\cong\mathbb{Z}^2\oplus\mathbb{Z}/2$, $(1,3,1)$; $T^2\#\mathbb{R}P^2\cong\#^3\mathbb{R}P^2$.]

10. For the word $a\,b\,c\,a^{-1}b^{-1}c^{-1}$, find the vertex classes, χ, the surface, and its Betti numbers over $\mathbb{Z}/2$. **[Not on Midterm 1]** Why can Proposition 9.5 not be applied directly?
    [Two classes $\{P_0,P_2,P_4\}$ and $\{P_1,P_3,P_5\}$; $\chi=2-3+1=0$, orientable: the torus, $b=(1,2,1)$. $H_1\cong\mathbb{Z}^2$, while the naive recipe would give $\mathbb{Z}^3$.]

11. **[Not on Midterm 1]** Orient the triangles of the 5-vertex Möbius band as $[1,2,3]$, $[2,3,4]$, $[3,4,5]$, $[1,4,5]$, $[1,2,5]$ and compute ∂_2 of their sum. Deduce that the boundary loop $1-3-5-2-4-1$ is homologous to twice the core loop $1-2-3-4-5-1$. Conclude that there is no retraction of the Möbius band onto its boundary circle.

12. Let K consist of the boundary of a tetrahedron and one extra edge $\{1,5\}$ from a vertex 1 of the tetrahedron to a new vertex 5. Compute the Betti numbers of K and the link of the vertex 1. **[Not on Midterm 1]** Is $|K|$ a surface?
[$b=(1,0,1)$; lk(1) is a hollow triangle together with the isolated vertex 5, $b=(2,1)$; no.]

13. True or false, with a reason: (a) If $b_0(K)=1$ and $b_k(K)=0$ for all $k\geq 1$, then K is a cone. (b) If $VR_\epsilon(X)$ has $b_2\neq 0$, then X is not contained in a line. (c) For every finite $X\subset\mathbb{R}^2$ and every ϵ, $b_2(\check{C}_\epsilon(X))=0$.
[(a) False: the strip of triangles $\{1,2,3\},\{2,3,4\},\{3,4,5\},\{4,5,6\}$ is a triangulated disk with $b=(1,0,0)$, but no vertex lies in all four triangles, so it is not a cone. (b) True: for X on a line, VR_ϵ and $\check{C}_\epsilon$ coincide and are nerves of intervals, which have the homology of a union of intervals. (c) True: by the Nerve Theorem it has the homology of a union of disks in the plane, which has no 2-dimensional homology.]

14. **(Programming.)** Write a function that takes a list of maximal simplices and returns the Betti numbers over $\mathbb{Z}/2$, by building the boundary matrices and computing their ranks by Gaussian elimination with XOR. Test it on all examples of these notes, including the hexagon of Example 8.1.